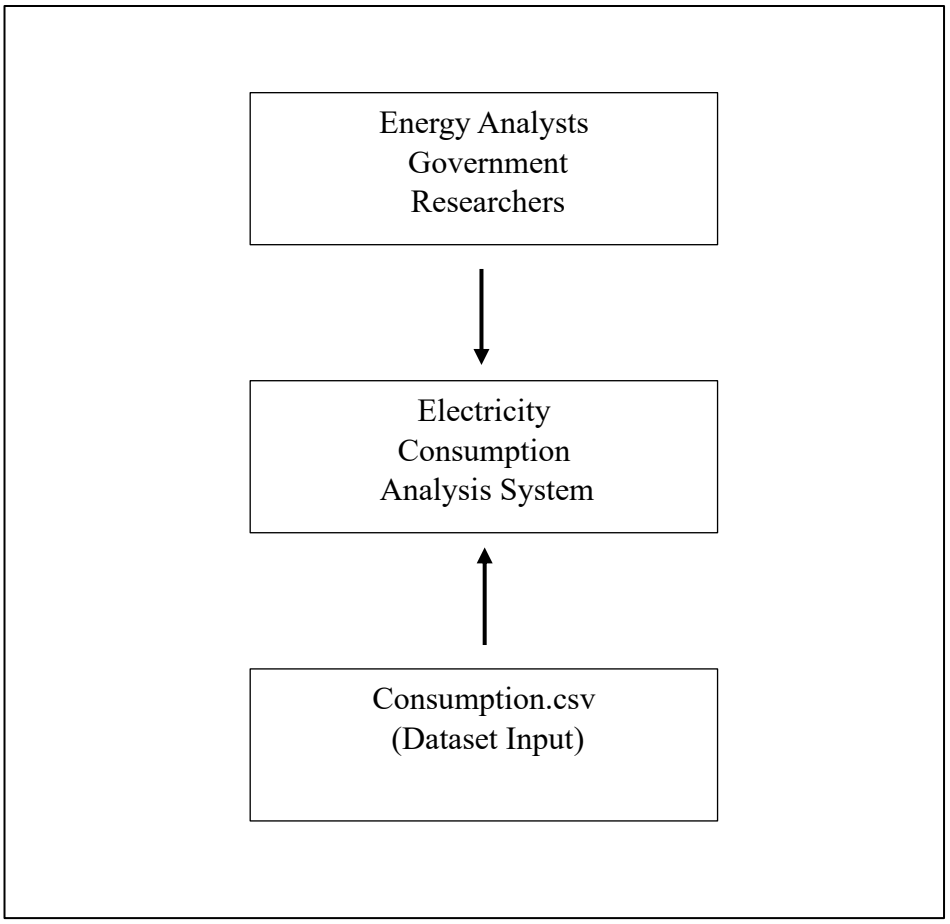
Project Design Phase-II

Data Flow Diagram & User Stories

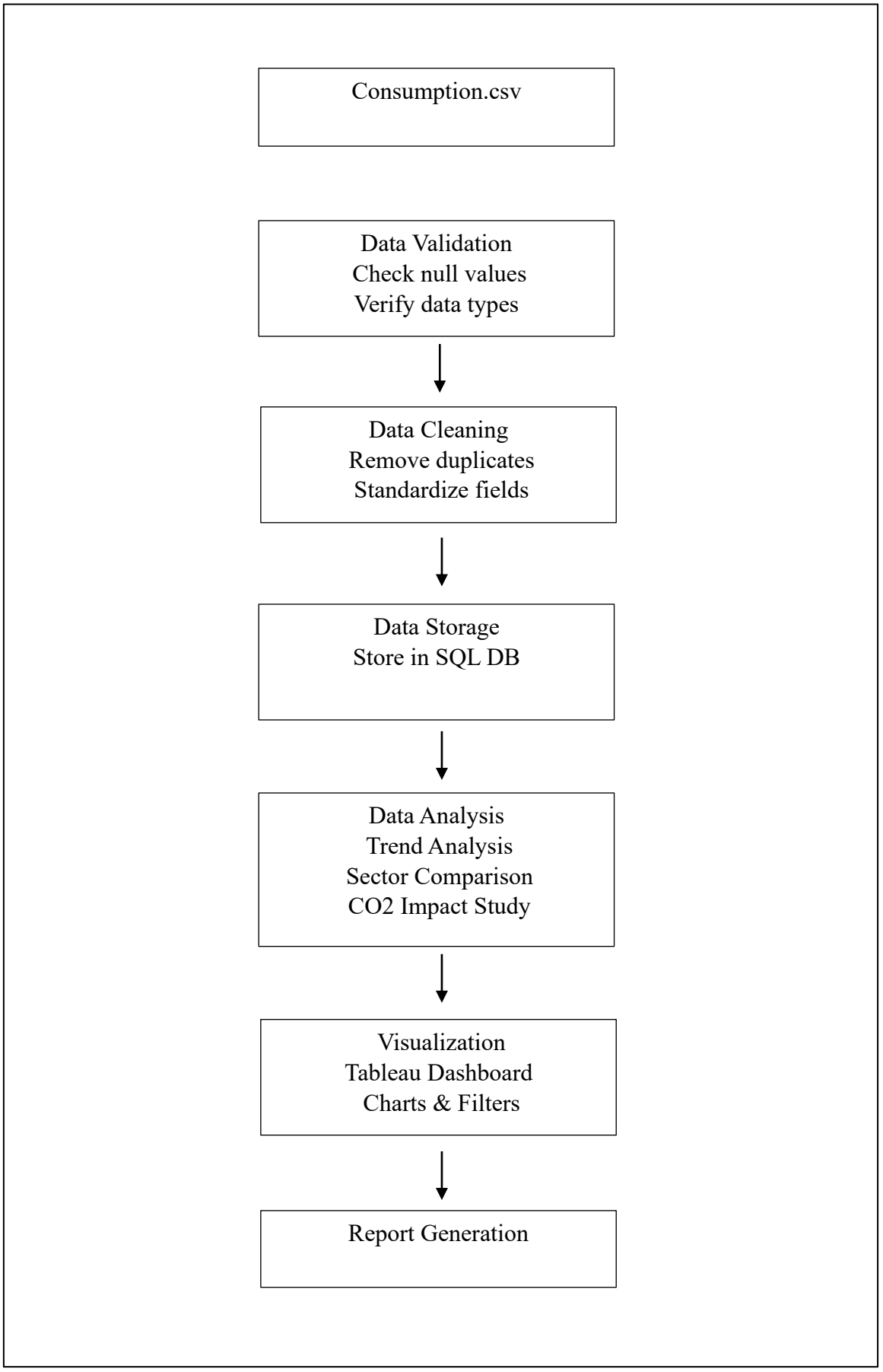
Date	20 February, 2026
Team ID	Rohit Sutar
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Data Flow Diagrams:

- DFD – Level 0 (Context Diagram): -



- **DFD – Level 1 (Major Processes): -**



DFD – Level 2 (Detailed Technical Flow): -

```
1. Data Input Module
   ├── Upload CSV
   ├── Schema Verification
   └── Field Mapping

2. Preprocessing Module
   ├── Null Handling
   ├── Outlier Detection
   ├── Data Type Conversion
   └── Formatting (Uppercase, Trim)

3. Database Layer
   ├── Insert Clean Data
   ├── Indexing
   └── Query Execution

4. Analytics Engine
   ├── Year-wise Aggregation
   ├── Sector-wise Aggregation
   ├── Region-wise Analysis
   ├── Renewable vs Non-Renewable Comparison
   └── CO2 Correlation Calculation

5. Visualization Layer
   ├── Time Series Chart
   ├── Bar Chart
   ├── Pie Chart
   ├── Map Visualization
   └── Scatter Plot

6. User Interaction Layer
   ├── Filter by Year
   ├── Filter by Sector
   ├── Filter by Region
   └── Export Report
```

User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority
User Story ID	**User Role**	**User Story**	**Acceptance Criteria**	**Priority**	High
US-1	Energy Analyst	As an Energy Analyst, I want to view yearly electricity consumption trends so that I can analyze growth patterns and forecast future demand.	Line chart displays Year vs Consumption accurately with filter options.	High	High
US-2	Government Policy Maker	As a Policy Maker, I want to compare renewable and non-renewable energy usage so that I can create sustainable energy policies.	System shows energy source comparison chart with clear data representation.	High	Low
US-3	Environmental Researcher	As a Researcher, I want to analyze the relationship between electricity consumption and CO_2 emissions so that I can assess environmental impact.	Scatter/analysis chart shows correlation between consumption and CO_2 emissions.	Medium	Medium
US-4	Regional Planner	As a Regional Planner, I want to view region-wise electricity consumption so that I can allocate energy resources efficiently.	Region-based visualization (map/bar chart) updates dynamically with filters.	High	High
US-5	Business Stakeholder	As a Business Stakeholder, I want to compare sector-wise electricity consumption so that I can understand sector demand patterns.	Sector comparison bar chart is available with sorting/filter options.	Medium	
US-6	Administrator	As an Administrator, I want to upload and manage electricity consumption datasets so that the system always uses updated and accurate data.	Admin can upload, validate, and update datasets successfully.	High	